

Assignment 1

◆ Task 1: Data Entry & Variable Setup

Create a dataset with at least 10–15 observations using the following variables:

- Name (String)
- Age (Numeric)
- Gender (Nominal)
- Department (Nominal)
- CGPA (Scale)

 Requirements:

- Define variables in Variable View
- Set correct variable types
- Add labels & value labels
- Define missing values (if applicable)
- Enter all data in Data View

Variable View:

[illegible]

Data View:

Visible: 5 of 5 Variables																	
	Name	Age	Gender	Department	CGPA	var	var	var	var	var	var	var	var	var	var	var	var
1	Avif Hasan	22	Male	Statistics	3.42												
2	Nusrat Jahan	27	Female	Computer ...	3.78												
3	Tamir Ahm...	19	Male	Business ...	3.15												
4	Sadia Rah...	31	Female	Economics	3.56												
5	Mehedi Ha...	24	Male	Mathematics	3.89												
6	Farzana A...		Female	English	3.33												
7	Rakib Hos...	35	Male	Statistics													
8	Tanjim Islam	21	Male	Software E...	3.61												
9	Jannatul F...	26	Female	Sociology	3.74												
10	Shakib Khan	33	Male	Marketing	3.21												
11	Mim Akter	20	Female	Psychology	3.47												
12	Fahim Cho...	28	Male	Finance	3.92												
13	Priya Sulta...	25	Female	Statistics	3.68												
14	Samuel Islam	30	Male	Managemen...	3.27												
15	Ayesha Noor	23	Female	Economics	3.84												
16																	
17																	
18																	
19																	
20																	
21																	
22																	
23																	
24																	
25																	
26																	

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	Name	String	25	0	Student Full Name	None	None	8	Left	Nominal	Input
2	Age	Numeric	8	0	Age of the Student (in Years)	None	None	8	Right	Scale	Input
3	Gender	String	8	0	Gender of the Student	{1, Male}...	None	8	Left	Nominal	Input
4	Department	String	31	0	Department of the Student	None	None	8	Left	Nominal	Input
5	CGPA	Numeric	8	2	Cumulative Grade Point Average (...)	None	None	8	Right	Scale	Input
6											
7											
8											
9											
10											
11											
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27											
28											

Value Labels

Value:

Label:

1 = "Male"

2 = "Female"

OK Cancel Help

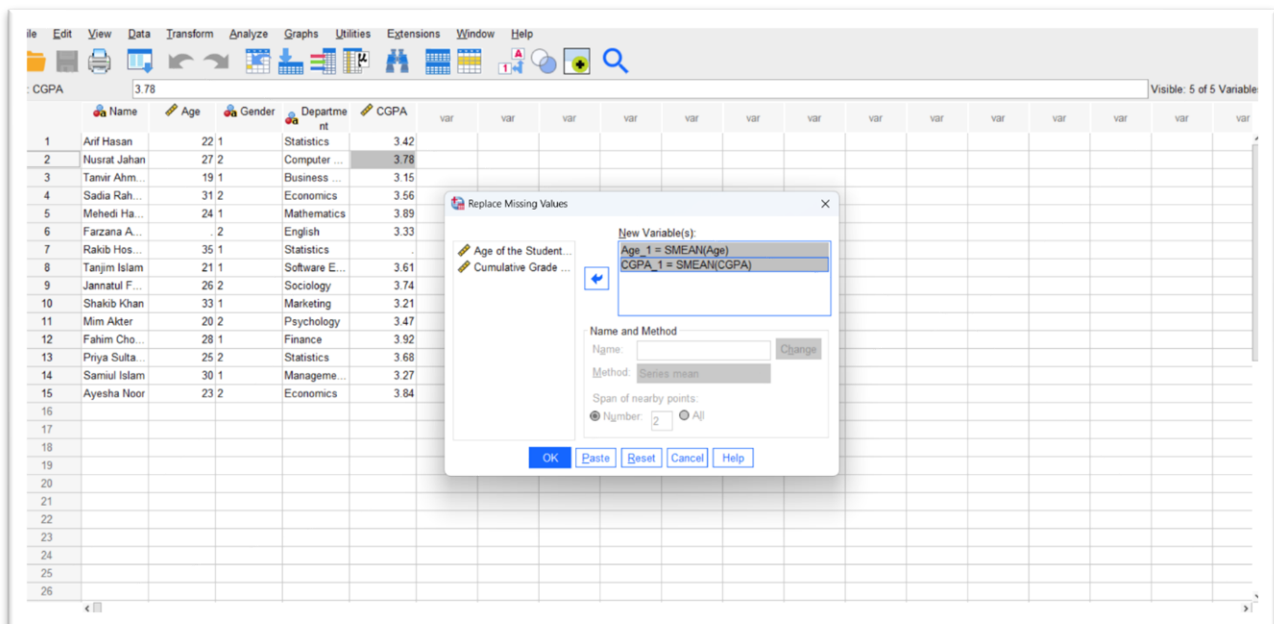
◆ Task 2: Data Cleaning

Perform the following:

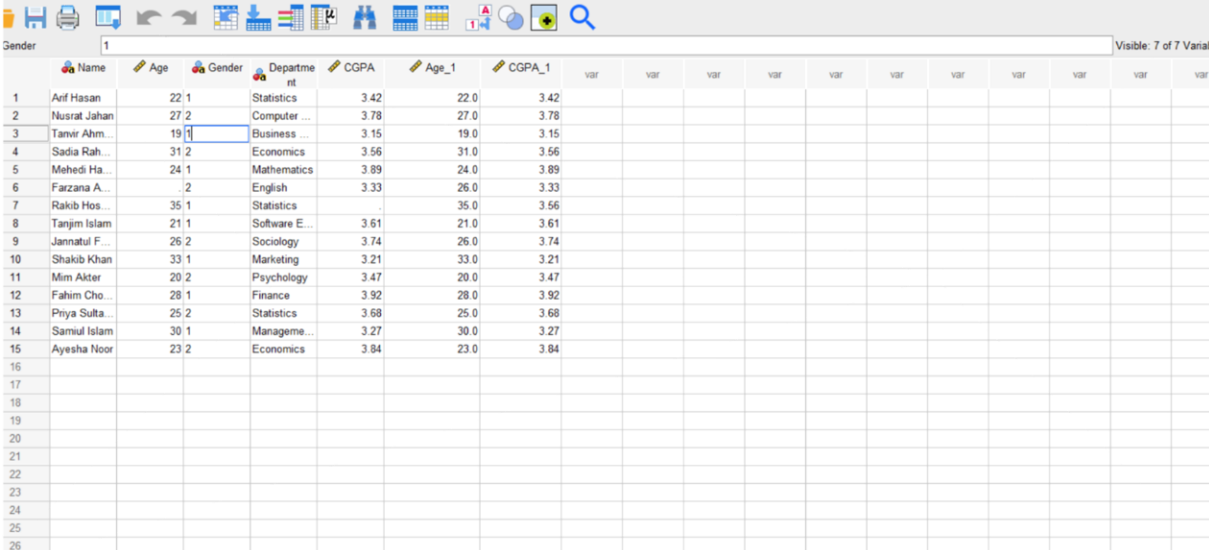
- Identify at least one missing value
- Handle the missing value properly
- Use Recode (same or different variable)
 - Example: Gender → Male = 1, Female = 2
- Create at least one new variable (using Recode/Compute)

Statistics						
		Student Full Name	Age of the Student (in Years)	Gender of the Student	Department of the Student	Cumulative Grade Point Average (CGPA)
N	Valid	15	14	15	15	14
	Missing	0	1	0	0	1

Interpretation: We can see from the frequency table that there is one missing value in both the Age and CGPA variables



Result Variables						
	Result Variable	N of Replaced Missing Values	Case Number of Non-Missing Values		N of Valid Cases	Creating Function
			First	Last		
1	Age_1	1	1	15	15	SMEAN(Age)
2	CGPA_1	1	1	15	15	SMEAN(CGPA)



	Name	Age	Gender	Department	CGPA	Age_1	CGPA_1	var	var	var	var	var	var	var	var	var	var
1	Arif Hasan	22	1	Statistics	3.42	22.0	3.42										
2	Nusrat Jahan	27	2	Computer ...	3.78	27.0	3.78										
3	Tanvir Ahm...	19	1	Business ...	3.15	19.0	3.15										
4	Sadia Rah...	31	2	Economics	3.56	31.0	3.56										
5	Mehedi Ha...	24	1	Mathematics	3.89	24.0	3.89										
6	Farzana A...	2	2	English	3.33	26.0	3.33										
7	Rakib Hos...	35	1	Statistics	.	35.0	3.56										
8	Tanjim Islam	21	1	Software E...	3.61	21.0	3.61										
9	Jannatul F...	26	2	Sociology	3.74	26.0	3.74										
10	Shakib Khan	33	1	Marketing	3.21	33.0	3.21										
11	Mim Akter	20	2	Psychology	3.47	20.0	3.47										
12	Fahim Cho...	28	1	Finance	3.92	28.0	3.92										
13	Priya Sulta...	25	2	Statistics	3.68	25.0	3.68										
14	Samiul Islam	30	1	Manageme...	3.27	30.0	3.27										
15	Ayesha Noor	23	2	Economics	3.84	23.0	3.84										
16																	
17																	
18																	
19																	
20																	
21																	
22																	
23																	
24																	
25																	
26																	

Interpretation: The dataset initially contained one missing value in both the Age and CGPA variables. To handle this problem properly, the missing values were replaced using the mean substitution method. After replacement, two new variables named Age_1 and CGPA_1 were created to store the cleaned values. This process improved the quality and completeness of the dataset, making it suitable for further statistical analysis.

Interpretation: The new variable ****Performance Level**** was created from the CGPA variable to categorize students based on their academic performance. Students with lower CGPA were classified as “Low,” moderate CGPA as “Moderate,” and higher CGPA as “High.” This transformation makes the data easier to interpret and helps compare student performance more effectively during analysis.

◆ Task 3: Descriptive Statistics

Generate:

- Frequency table (Gender / Department)
- Mean, Median, Mode (Age / CGPA)
- Standard Deviation

◆ Task 4: Interpretation

- What is the average age?
- Which gender appears most frequently?
- What does standard deviation indicate?

Gender of the Student					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	8	53.3	53.3	53.3
	Female	7	46.7	46.7	100.0
	Total	15	100.0	100.0	

Department of the Student					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Business Administration	1	6.7	6.7	6.7
	Computer Science & Engineering	1	6.7	6.7	13.3
	Economics	2	13.3	13.3	26.7
	English	1	6.7	6.7	33.3
	Finance	1	6.7	6.7	40.0
	Management Information Systems	1	6.7	6.7	46.7
	Marketing	1	6.7	6.7	53.3
	Mathematics	1	6.7	6.7	60.0
	Psychology	1	6.7	6.7	66.7
	Sociology	1	6.7	6.7	73.3
	Software Engineering	1	6.7	6.7	80.0
	Statistics	3	20.0	20.0	100.0
	Total	15	100.0	100.0	

Statistics			
		SMEAN(Age)	SMEAN(CGPA)
N	Valid	15	15
	Missing	0	0
Mean		26.000	3.5621
Median		26.000	3.5621
Mode		26.0	3.15 ^a
Std. Deviation		4.7509	.24926
a. Multiple modes exist. The smallest value is shown			

Task 4: Interpretation

- The average age of the students is 26 years. This means most students in the dataset are around 26 years old on average.
- From the frequency table, the Male gender appears most frequently in the dataset.
- The standard deviation shows how much the data values vary from the mean.
 - The standard deviation of Age (4.7509) indicates that students' ages are moderately spread around the average age.
 - The standard deviation of CGPA (0.24926) indicates that most CGPA values are close to the mean CGPA, meaning there is low variation in academic performance among students.